Text-to-Image Activation for Open-Vocabulary Semantic Segmentation in Remote Sensing

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***Abstract*—Open-vocabulary semantic segmentation (OVSS) in remote sensing aims to recognize arbitrary object categories from satellite imageries beyond a fixed label set, but its progress is constrained by the lack of fine-grained pixel-level supervision during training. Text-based weak supervision offers a scalable alternative by generating pseudo-labels from image-level annotations, yet in remote sensing scenarios it faces twofold challenges: 1) incompleteness and unreliability of pseudo-labels and 2) segmentation learning degradation induced by label noise. To address these issues, we propose a framework named Text-to-Image Activation for OVSS (T2ASeg), tailored for the spatial and semantic characteristics of satellite imagery. T2ASeg integrates two key modules to enhance pseudo-label quality and segmentation robustness. The text-guided adaptive activation module (TAAM) generates scale-aware and semantically consistent class activation maps (CAMs), serving as reliable spatial and semantic cues for pseudo-label generation. The multiscale structure and semantic enhancement module (MSSEM) combines structural modeling with class-aware semantic guidance, enabling the model to better capture fine-grained semantics and spatial structures under noisy supervision. By synergizing these modules, T2ASeg delivers more accurate and complete segmentation maps, even in the presence of weak and noisy labels. Extensive experiments on multiple remote sensing benchmarks show that T2ASeg outperforms state-of-the-art methods in both accuracy and boundary quality, demonstrating the value of combining textual guidance with adaptive spatial modeling for open-vocabulary segmentation. Code is available at https://github.com/WyneeWang/Access-Instructions-for-T2ASeg.**

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I. Introduction

In THE field of semantic segmentation for remote sensing images, real-world scenarios involve diverse, fine-grained, and dynamically evolving land cover categories, which makes traditional closed-set methods, trained on fixed label systems with costly pixel-level annotations, difficult to scale [1]. To address this limitation, open-vocabulary semantic segmentation (OVSS) has recently emerged as a promising paradigm [2]. Unlike closed-set approaches that are restricted to predefined categories, OVSS aims to recognize arbitrary categories, including those unseen during training, by leveraging semantic information encoded in natural language. This paradigm shift enables models to generalize beyond fixed taxonomies and provides stronger adaptability to large-scale and heterogeneous remote sensing data. However, training OVSS models typically requires high-quality pixel-level annotations, and the scarcity of such expert-labeled data constitutes a major obstacle in practice. To mitigate this challenge, recent studies have explored text-based weakly supervised methods, in which image-level textual annotations and vision-language alignment facilitate learning category-aware representations and generating pseudo-labels as substitutes for dense annotations [3], [4], [5].

Pseudo-labels are central to the weakly supervised segmentation pipeline, as their quality directly impacts the model's ability to recognize semantic categories and localize them accurately in space. Against this background, this work investigates OVSS in remote sensing under text-based weak supervision, aiming to design an efficient text-guided pseudo-label generation strategy that enables fine-grained recognition of open-set categories without relying on pixel-level annotations. However, this task poses twofold challenges: 1) incompleteness and unreliability of pseudo-labels and 2) segmentation learning degradation induced by label noise.

To elaborate on the first challenge, we examine how current pseudo-label generation methods suffer from incomplete and unreliable activation, particularly in the context of class activation map (CAM) approaches. CAM identifies image regions that are most relevant to specific class semantics, thereby providing an initial mask for pseudo-label generation. A typical example is the Grad-CAM-based method [6]. As illustrated in

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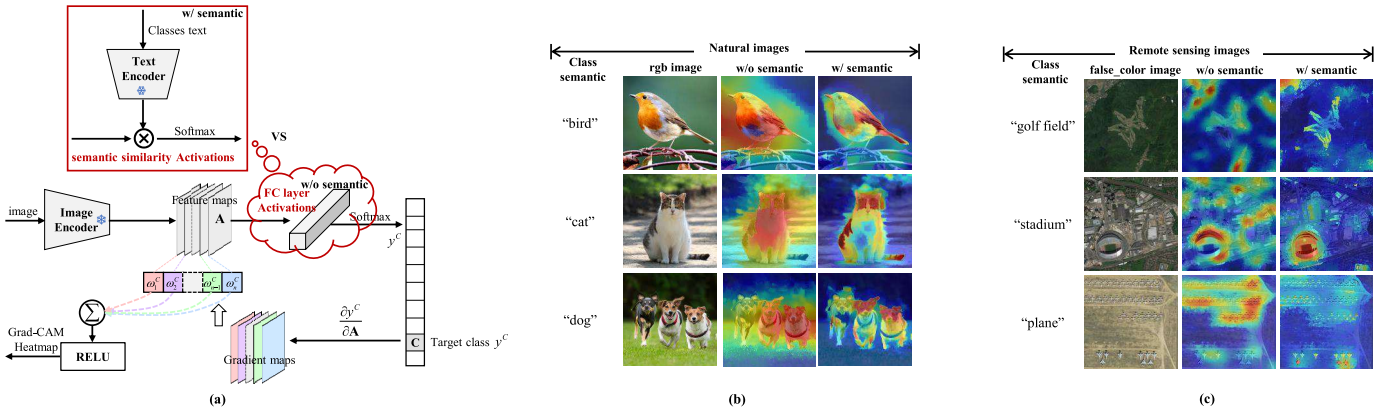


Fig. 1. Illustration of the class activation mechanism: (a) CAM generation based on Grad-CAM, where semantic similarity activation is adopted to replace the conventional FC-layer activation under text-supervised settings; (b) examples of CAMs on natural images; and (c) examples of CAMs on remote sensing images.

Fig. 1(a), Grad-CAM generates activation maps by computing the gradients of prediction scores with respect to visual features, and then weighting these gradients over the feature maps. The resulting CAM highlights regions that contribute most to the model's classification decision. However, such methods rely heavily on a fixed set of categories defined by the training dataset. As shown in the "w/o semantic" column in Fig. 1(b), the model outputs a distribution over predefined categories and emphasizes only the regions important for classification, limiting their applicability to open-vocabulary scenarios.

In contrast, image-level textual descriptions offer a more flexible and expressive form of supervision for open-set recognition. Unlike traditional classification labels constrained to closed vocabularies, text annotations can describe a wide range of semantic categories while also providing richer guidance. Through vision-language alignment, models can compute similarity between image features and textual embeddings to generate class-relevant activation maps, thereby improving the semantic focus of CAM. This idea has been widely adopted to enhance CAM quality. Yet, vision-language models like CLIP [7] are primarily pretrained for global semantic discrimination, not for precise regional alignment. Since their training aligns whole-image representations with sentence-level descriptions, they often overlook fine-grained spatial details. To address this, Cha et al. [8] proposed a masking mechanism that limits attention to target regions during training, promoting better region-text alignment. Similarly, SegEarth-OV [9] removes the classification token from the visual features to suppress global semantic interference. As shown in the "w/ semantic" column of Fig. 1(b), these improvements yield activation maps that are more focused and better aligned with object regions.

Despite these advances, a fundamental limitation remains: CAM-based approaches tend to highlight only the most discriminative parts of an object, often failing to capture its complete spatial extent [10], [11]. This issue is especially pronounced in remote sensing imagery. As illustrated in Fig. 1(c), objects in remote sensing images often have abstract semantics and weak visual cues, such as different vegetation types may appear visually similar. The semantic gap between text and image makes region activation more difficult. In addition, remote sensing data typically contain diverse object scales, with small targets and large land cover regions coexisting, along with irregular shapes and blurred textures. These factors pose significant challenges for vision-language models to accurately localize small or subtle structures. As a result, weak or missing activations frequently occur, particularly at fine-grained levels. Therefore, in the context of text-based weak supervision, one key challenge lies in how to generate activation maps that are not only semantically consistent, but also spatially complete and structurally coherent.

To elaborate on the second challenge, we focus on the segmentation learning issues caused by label noise under image-level weak supervision. Without pixel-level annotations during training, the generated pseudo-labels often contain semantic bias and labeling errors, commonly referred to as noisy pseudo-labels [12]. These errors introduce incorrect supervision signals that mislead the model, causing it to learn suboptimal or even wrong representations and ultimately reducing segmentation accuracy.

On one hand, semantic uncertainty in pseudo-labels misguides the learning of class-discriminative features, reducing the model's ability to distinguish fine-grained semantic differences. On the other hand, spatial misactivations and boundary ambiguity compromise the structural integrity of segmented objects. This problem is particularly pronounced in remote sensing imagery, where land cover types such as roads, buildings, and water bodies often appear adjacent to each other, share similar textures, or exhibit fuzzy boundaries, making the model more prone to confusion when trained with noisy supervision.

Moreover, the open-vocabulary nature of the task exacerbates this issue. Since the model must generalize to unseen or dynamically emerging categories without direct visual samples, it heavily depends on the accuracy and reliability of pseudo-labels. When these labels are noisy, the resulting erroneous supervision undermines the model's generalization capability, leading to unstable or inconsistent predictions for novel classes. Therefore, in weakly supervised OVSS for remote sensing, a key challenge is to mitigate the negative

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impact of noisy pseudo-labels on both semantic understanding and structural consistency, ensuring effective learning and generalization in complex open-world scenarios.

Therefore, to address the above challenges, we propose T2ASeg, a novel framework named text-to-Image activation for OVSS, tailored for remote sensing scenarios. The core idea is to bridge text semantics and image regions through adaptive activation while reinforcing the model's ability to resist noise during segmentation learning. T2ASeg adopts a dual-stage design consisting of pseudo-label generation and decoding enhancement, with each stage directly addressing a corresponding challenge. First, to alleviate the incompleteness and semantic inconsistency in pseudo-label generation, we improve a text-guided adaptive activation module (TAAM) that adaptively produces category-aware activation maps. Based on these high-quality activation maps, spatially complete and semantically aligned pseudo-labels are generated, which are especially effective for recognizing fine-grained and small-scale objects in complex remote sensing scenes. Second, to mitigate the adverse effects of noisy pseudo-labels during training, we design a structure-semantic collaborative enhancement strategy in the decoding stage. This strategy integrates multiscale structural cues and category-guided semantic priors to improve the model's discrimination ability around ambiguous boundaries and confusing classes, thereby enhancing its robustness and generalization capability under weak supervision and open-vocabulary settings. The main contributions of this work are summarized as follows.

1) A text-guided adaptive activation method is proposed, utilizing an anti-symmetric design that integrates coarse-grained saliency activation (CSA) with fine-grained semantic guidance. This captures spatial responses across diverse object scales. A confidence-aware fusion strategy further adapts the method to the varied scale distributions inherent in remote sensing land cover.
2) A visual feature-driven self-attention refinement mechanism is introduced to enhance consistent activation within intra-class regions. This mitigates activation noise and semantic drift arising from imperfect image–text alignment, improving pseudo-label accuracy and semantic stability.
3) A multiscale structure and semantic enhancement module (MSSEM) is designed for the decoding phase. By integrating structural modeling and semantic alignment, this module enhances robustness against noisy supervision and significantly improves semantic parsing capability and segmentation accuracy under open-vocabulary conditions.

The remainder of this article is organized as follows. Section II reviews related work. Section III presents the proposed methodology. Section IV reports experimental results. Section V offers discussion. Section VI concludes this article.

II. Related Work

A. OVSS in Remote Sensing

In remote sensing image analysis, land-cover categories are numerous and unevenly distributed, and a large number of open-vocabulary classes that do not appear in the training set are often encountered in real-world applications. The key requirement for improving the intelligent interpretation capability of remote sensing imagery is to identify both seen and unseen classes within a unified semantic space while producing high-precision pixel-level segmentation results. Due to the ultrahigh resolution of remote sensing imagery, the complex and diverse morphology of land-cover objects, sparse semantic distribution, and ambiguous boundaries, traditional fully supervised segmentation methods that rely on large-scale pixel-level annotations are difficult to generalize. In recent years, open-vocabulary approaches that integrate visual and textual information have provided new solutions to this problem. Among them, text-weak supervision, which can be trained using only image–text pairs as weak labels, has shown great potential in the OVSS tasks.

Existing studies on text weakly supervised OVSS have mainly focused on natural images and can be broadly categorized into two types. The first type emphasizes region consistency modeling, which enhances cross-modal alignment by improving the semantic consistency between local image regions and textual labels. For example, GroupViT [13] leverages the grouping structure of vision transformers to achieve bottom-up region aggregation and alignment; PGSeg [14] introduces nonlearnable prototypes to improve semantic consistency; OVSegmentor [15] applies slot-attention to group pixels into semantic entities and incorporates entity completion and cross-image consistency tasks to enhance the generalization ability to unseen classes; SimSeg [16] proposes the LoDA strategy to strengthen local feature alignment, addressing the limitation of CLIP in modeling only the global image; and SegCLIP [17] combines learnable aggregation centers with superpixel regularization to improve structural awareness. The second type focuses on text-guided pseudo-label generation, which produces initial masks under textual prompts and progressively refines them to replace costly manual annotations. For example, MaskCLIP [18] directly uses CLIP-generated CAMs as initial masks and optimizes segmentation performance through self-training; WLSegNet [19] leverages class context matching to improve generalization in zero- and few-shot settings; S-Seg [20] jointly trains pseudo-masks and text to enable cross-dataset transfer; Unpair-Seg [21] fuses supervisory signals from image–text pairs and image–mask pairs to mitigate mismatches between masks and semantic entities; and TCL [8] combines pseudo-label generation with region-level alignment, simultaneously improving label quality and structural alignment.

However, these methods often assume that target structures are continuous and boundaries are well-defined, which limits their applicability to remote sensing images that commonly exhibit complex object morphology, blurred boundaries, and sparse semantic distribution. Such characteristics can lead to unstable cross-modal alignment and inaccurate object localization. In comparison, text-guided pseudo-label generation is more promising in remote sensing scenarios, as it can exploit high-resolution visual features, multiscale fusion mechanisms, and self-training optimization to progressively improve pseudo-label quality while enhancing the segmentation of

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open-vocabulary classes. Therefore, effectively integrating the high-resolution characteristics of remote sensing imagery with rich textual semantic priors to generate structurally coherent and semantically accurate pseudo-labels is a key direction for improving OVSS performance in complex land-cover scenes.

B. Pseudo-Labeling in Weakly-Supervised Segmentation

With the advent of vision-language pretrained models, strong cross-modal alignment capabilities have enabled pseudo-label generation from image-level annotations to become a prominent paradigm in weakly supervised semantic segmentation. These methods typically establish correspondence between images and category descriptions via vision–text alignment, thereby generating semantic region masks to replace manual annotations. Some works aim to improve the generalization and robustness of pseudo-labels by enhancing semantic discriminability through strategies such as prototype-based contrastive learning [22], class-aware augmentation [23], and multimodal feature fusion [24], [25]. Others focus on controlling mask quality, employing techniques like noise suppression, self-training with distillation, or localization-guided refinement to improve pseudo-label reliability [26], [27].

Among the various pseudo-labeling strategies, CAM remains one of the most widely adopted tools. CAM identifies regions in the image that are most relevant to the category semantics, serving as an initial mask for pseudo-label generation. However, traditional CAM variants, such as GradCAM, ScoreCAM, and LayerCAM, tend to highlight only the most discriminative parts of objects, often failing to cover the full spatial extent of the target while introducing background noise, thereby compromising mask completeness and structural integrity. To address these issues, recent studies have proposed enhancements to the CAM mechanism. For example, OAM-Net [28] employs an object-aware structure with multiple loss functions to ensure activation region completeness and consistency. CPMC [29] introduces a dual-branch architecture combining classifier-based and prototype-based CAMs, incorporating a context-dependent calibration mechanism. Context-CAM [30] further integrates a context enhancement module and a semantic reverse fusion mechanism to alleviate under-activation in nondiscriminative regions. In addition, saliency guidance and attention mechanisms have been utilized to improve boundary sensitivity. Notably, saliency-guided Transformer CAM [31] and WaveCAM [32] propose innovations in attention modeling and semantic decoupling of foreground/background content, respectively, yielding more complete pseudo-label responses. Even in specialized domains such as remote sensing, methods like water-matching CAM [33] incorporate prior physical characteristics and consistency modeling, significantly improving pseudo-label accuracy and adaptability in SAR imagery.

Despite notable advances in natural image settings for open-vocabulary and weakly supervised segmentation, challenges persist when tackling remote sensing scenarios characterized by both open vocabularies and image-level supervision constraints. Structural incompleteness, semantic misalignment, and limited generalization are still prevalent. On one hand, many existing methods rely on complex training procedures or mask refinement strategies, which are often infeasible in remote sensing tasks where labeled data is scarce, category sets are open-ended, and object scales vary drastically. On the other hand, most designs are tailored to the texture structures and object saliency distributions typical of natural images, leading to misaligned activations and background noise when transferred to remote sensing imagery. Hence, a key research direction is to develop pseudo-labeling mechanisms that are training-free, capable of integrating multisource activation signals, and inherently compatible with the characteristics of remote sensing imagery, all built upon the strong vision-language alignment of pretrained models. Such approaches hold the potential to mitigate issues like semantic drift, incomplete responses, and structural incoherence under weak supervision, thereby advancing OVSS in complex remote sensing environments.

III. Proposed Method

Fig. 2 illustrates the proposed T2ASeg framework tailored for open-vocabulary remote sensing. T2ASeg is designed around two critical stages: pseudo-label generation and decoder enhancement, in which a TAAM and a MSSEM are introduced to enable high-quality pseudo-labeling and robust semantic decoding under weak supervision. During the pseudo-label generation stage, TAAM leverages image-level textual annotations to guide a dual-branch structure that fuses CSA and fine-grained semantic guidance, allowing for adaptive and precise localization of multiscale target regions. The fused activation maps are then refined through a self-attention mechanism based on visual features, which enhances both semantic coherence and spatial integrity of the pseudo-labels, and mitigates activation noise caused by text-image misalignment. In the decoder enhancement stage, MSSEM jointly incorporates multiscale structural modeling and text-guided semantic enhancement, improving the decoder's ability to resist pseudo-label noise and to reason over spatial semantics. This strengthens the model's segmentation accuracy and generalization capability under open-vocabulary settings.

A. Feature Embedding

In the proposed framework, the input includes remote sensing images and their corresponding candidate category text labels, which are extracted using pretrained image encoder and text encoder, respectively. We select the CLIP model, which has superior visual-language alignment performance, and use its pretrained ViT/B-16 [34] architecture as the encoders. Given an input image $I \in \mathbb{R}^{H\times W\times 3}$ of $H \times W$ size, the CLIP image encoder Φ^I produces multilayer feature representations across its transformer blocks. Let $\Phi^I(I) = \{f_l^I\}_{l=1}^L$ be the feature maps extracted from the ith block of the image encoder, where L is the total number of blocks. The final layer output $f_L^I \in \mathbb{R}^{(h\times w)\times d_I}$ is passed through a linear projection layer to produce the main image feature $\mathcal{E}^I \in \mathbb{R}^{(h\times w)\times d}$, where (h, w) is the spatial resolution and d_I, d denotes the channel dimensions before and after projection, respectively. In addition, a set of low- and mid-level features $\mathcal{E}^I_{\text{guide}} = \{f_s^I\}_{s\in S}$ is selected

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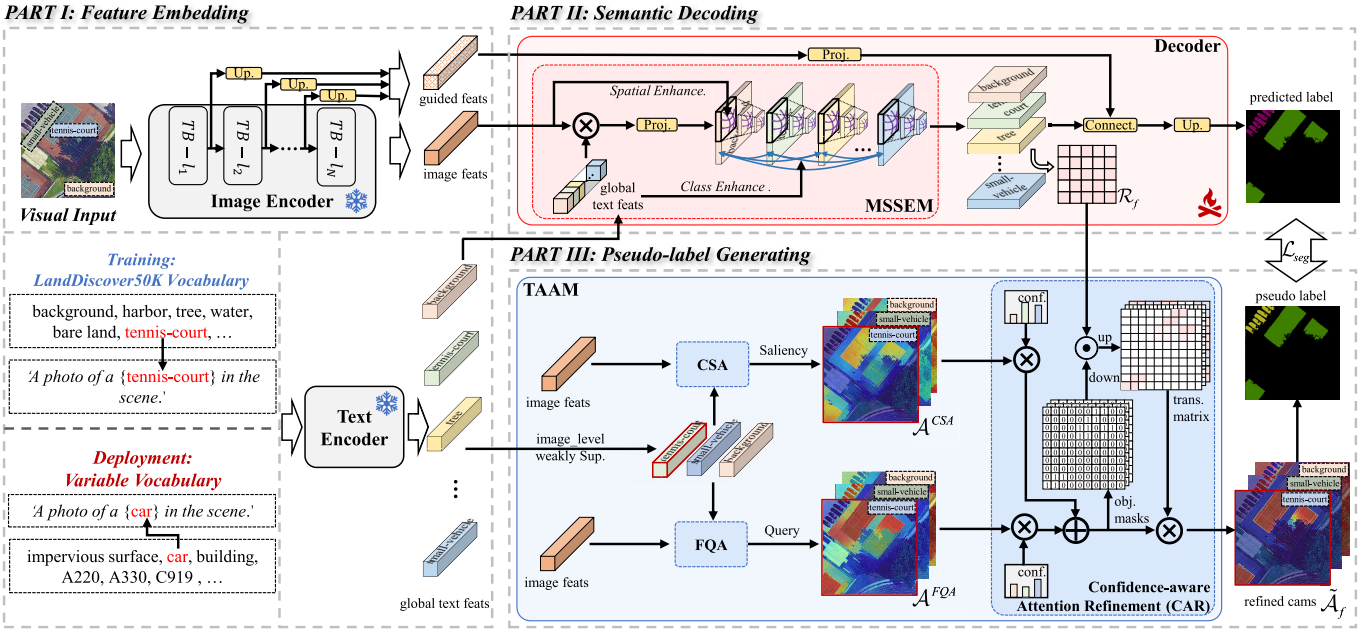


Fig. 2. Overview of the T2ASeg framework. The architecture consists of three key components: feature embedding, where the input image and class-level text labels are encoded by pretrained CLIP visual and text encoders; semantic decoding, where visual features are aligned with global textual embeddings to produce initial segmentation predictions, which are progressively refined and upsampled into the final segmentation map; and pseudo-label generation, where image-level textual labels are leveraged to compute text-guided adaptive CAMs from visual features, providing pseudo-labels that supervise the training of the semantic decoder.

from earlier transformer blocks, where $S \subseteq \{1, 2, \ldots, L-1\}$ represents the index set of selected layers. These intermediate features serve as residual information during the upsampling phase of the decoder and are fused with high-level semantic features to enhance boundary delineation and structural restoration in the final segmentation outputs. Simultaneously, the CLIP text encoder Φ^{T} processes the set of C candidate category labels $\{T_j\}_{j=1}^{C}$, generating the corresponding global class semantic representations $\mathcal{E}^{\mathrm{T}} = \Phi^{\mathrm{T}}(T) \in \mathbb{R}^{C\times d}$. These textual embeddings are later utilized in both image–text alignment and activation guidance throughout the pseudo-label generation and semantic decoding stages.

B. Semantic Enhancement and Residual Decoding

To address the class-agnostic segmentation requirement in open-vocabulary settings, our decoding stage utilizes an image–text similarity-driven cost volume map constructed via feature alignment between visual and textual representations. This design enables semantic-guided segmentation prediction under weak supervision. However, the use of text-level supervision may introduce semantic bias and spatial noise in the pseudo-labels, potentially degrading the decoder's discriminative capability and structural recovery accuracy. To mitigate this, we propose MSSEM that integrates structural modeling and semantic alignment mechanisms into the decoding process. This module is designed to improve the model's robustness and generalization under noisy supervision by jointly enhancing local structural reasoning and global semantic consistency. Furthermore, during upsampling, shallow and mid-level features are incorporated as multiscale residuals to refine spatial boundaries and recover fine-grained object structures.

1) Multiscale Structure and Semantic Enhancement: To enhance both semantic consistency modeling and structure awareness under text-weak supervision, MSSEM introduces improved multiscale structure modeling and category-level semantic guidance mechanisms on top of the cost volume. These components strengthen local spatial parsing and global class alignment, respectively, which are crucial for reliable segmentation in open-vocabulary remote sensing imagery.

A. Cost volume encoding: Let $\mathcal{E}^{I} \in \mathbb{R}^{(h\times w)\times d}$ be the main image feature map obtained from the CLIP image encoder, and let $\mathcal{E}^{\mathrm{T}} \in \mathbb{R}^{C\times d}$ be the global semantic embeddings of the C candidate category labels extracted by the text encoder. We normalize both the image and text features and compute the cost volume using cosine similarity

$$\mathcal{M}(i,c) = \frac{\mathcal{E}^{I}(i)\cdot\mathcal{E}^{\mathrm{T}}(c)}{\|\mathcal{E}^{I}(i)\|\|\mathcal{E}^{\mathrm{T}}(c)\|}. \tag{1}$$

Here, $\mathcal{M}(i,c) \in \mathbb{R}^{(h\times w)\times C}$ reflects the semantic similarity between each spatial location in the image and each candidate class label, capturing how likely a given pixel belongs to a specific category. To enhance the nonlinear modeling capacity and enrich semantic representations, we apply a convolutional projection $\varphi_{7\times 7}$ with kernel size 7×7 to transform the cost volume into a latent semantic space

$$\mathcal{Z} = \varphi_{7\times 7}(\mathcal{M}) \in \mathbb{R}^{(h\times w)\times C\times D} \tag{2}$$

where D is the dimensionality of the latent space, and $\mathcal{Z}$ serves as the semantic input for subsequent structural modeling and semantic enhancement components.

B. Multiscale structural enhancement: The structural enhancement component focuses on modeling the spatial configuration of each category across the cost volume. The

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primary goal is to extract multiscale contextual structure from remote sensing imagery, where objects often vary significantly in size, shape, and distribution. Traditional methods based on fixed receptive fields often struggle to capture such diversity, especially under weak supervision. To address this, we introduce a multiwindow Swin Transformer [35] design for structural encoding. Multiple Swin Transformer blocks are deployed in parallel, each configured with a distinct window size to model local structure under varying receptive fields. These blocks alternate between regular and shifted windows, enabling nonoverlapping attention and context capture at multiple scales. For the spatial representation $\mathcal{Z}(:,c)$ of category c in the cost volume, the structural enhancement under window size w is formulated as

$$\mathcal{Z}'_w(:,c) = \Psi^{st}_w(\mathcal{Z}(:,c)) \tag{3}$$

where Ψ^{st}_w represents the Swin Transformer operation at window size w. Then, to integrate the multiscale outputs, we assign each scale a learnable fusion weight α_w and perform a weighted linear combination to generate the final structure-enhanced feature representation

$$\hat{\mathcal{Z}}(:,c) = \sum_{w\in\mathcal{W}} \alpha_w \cdot \mathcal{Z}'_w(:,c) \tag{4}$$

where $\mathcal{W}$ denotes the set of window sizes. This adaptive fusion design allows the model to dynamically adjust the importance of different receptive fields during training, enabling it to better capture category-specific structural variations across multiple spatial scales.

C. Classwise semantic consistency enhancement: While the aforementioned multiscale structural modeling enhances the model's ability to parse complex object structures in remote sensing imagery, its reliance on pseudo labels derived from image-level annotations can introduce semantic noise and category confusion. These issues often degrade class discrimination and blur intercategory boundaries, especially under weak supervision. To mitigate this, we adopt a classwise semantic consistency enhancement mechanism, designed to explicitly model semantic dependencies and co-occurrence patterns across categories. This mechanism performs context aggregation along the category dimension, allowing the model to learn contextualized class representations and improve robustness to semantic deviations in pseudo labels. Concretely, for each spatial position i, we extract its class-specific feature matrix $\hat{\mathcal{Z}}(i,:)$. This matrix is treated as a token sequence and fed into a category-wise Transformer module Ψ^{cs}

$$\tilde{\mathcal{Z}}(i,:) = \Psi^{\text{cs}}\left(\hat{\mathcal{Z}}(i,:)\right). \tag{5}$$

This operation aggregates semantic information across all classes at a given spatial location, yielding context-aware representations for each category. In doing so, the model captures interclass relationships, such as "buildings co-located with roads" or "farmland adjacent to water bodies," which are crucial for disambiguating class semantics and suppressing misclassifications induced by pseudo label noise. Importantly, to ensure scalability and generalization under open-vocabulary conditions, where the number and order of class labels may vary, we implement a linear Transformer without positional encoding. This design offers order invariance, efficient computation with linear complexity, and compatibility with varying class vocabularies, which are essential properties for semantic modeling in open-world scenarios. Finally, the enhanced classwise semantic features $\tilde{\mathcal{Z}} \in \mathbb{R}^{(h\times w)\times C\times D}$ are subsequently passed to the decoder, serving as refined semantic guidance that reinforces class consistency and improves robustness to noisy supervision during final segmentation prediction.

2) Residual Decoding: To restore the enhanced classwise semantic representations to high-resolution segmentation maps, we adopt a residual-guided decoding strategy with progressive upsampling, where semantic predictions are gradually recovered to the original image size. This design alleviates spatial information loss and enhances structural fidelity under weak supervision. Specifically, we incorporate a set of multilevel intermediate features $\mathcal{E}^I_{\text{guide}} = \{f^I_s\}_{s\in S}$, extracted from shallow and intermediate layers of the CLIP image encoder, to serve as residual guidance. These features provide high-frequency spatial details, which are often lost in the deeper layers due to aggressive downsampling. Integrating them into the decoding process helps recover fine-grained details and mitigate local ambiguity or texture smoothing caused by noisy pseudo labels. Formally, let $\tilde{\mathcal{Z}}$ denote the semantic-enhanced features output by MSSEM. At each decoding stage, we perform spatial resolution recovery via deconvolution operations $\varphi^{\mathcal{D}}(\cdot)$ on both $\tilde{\mathcal{Z}}$ and the corresponding residual feature, followed by channelwise concatenation and fusion via a convolutional decoding block $\phi(\cdot)$

$$\hat{\mathcal{Y}} = \phi\left(\text{Concat}\left(\varphi^{\mathcal{D}}\left(\tilde{\mathcal{Z}}\right), \varphi^{\mathcal{D}}\left(\mathcal{E}^I_{\text{guide}}\right)\right)\right) \tag{6}$$

where $\hat{\mathcal{Y}} \in \mathbb{R}^{(H\times W)\times C}$ denotes the final segmentation prediction across C classes for each spatial location. The residual guidance facilitates the integration of hierarchical spatial cues with semantic reasoning, enabling sharper boundaries, better localization, and improved segmentation quality in challenging open-vocabulary and weak-supervision settings.

C. Text-Guided Adaptive Activation

Under the constraint of text-based weak supervision, the proposed TAAM leverages the semantic correlation between image–text pairs to generate high-quality CAMs. These activation maps are used to generate pseudo-labels, which are then employed to supervise the segmentation network during training. TAAM is specifically tailored to the characteristics of remote sensing imagery, where object scales vary widely and semantic alignment between images and text is often weak. It consists of three submodules: CSA, fine-grained query activation (FQA), and confidence-aware attention refinement (CAR). Among them, CSA and FQA are designed in an antisymmetric manner to capture spatial responses at different object scales, as illustrated in Fig. 3. CAR further collaborates with them to produce activation regions that are structurally complete, semantically focused, and spatially consistent.

1) Coarse-Grained Saliency Activation: The CSA submodule aims to produce coarse but semantically robust CAMs from low-resolution, high-level semantic features, capturing the salient response regions of land-cover objects

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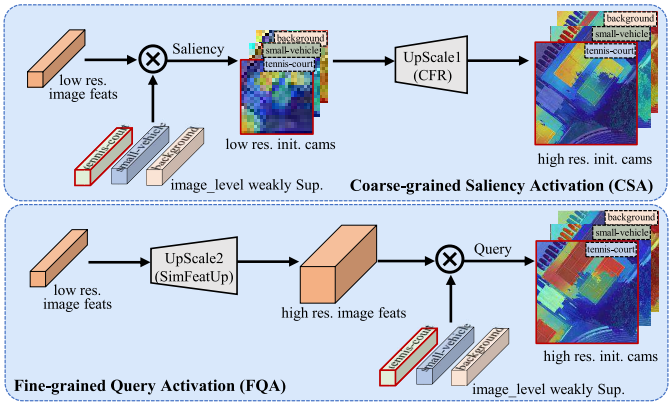


Fig. 3. Anti-symmetric design of CSA and FQA. CSA exploits low-resolution high-level semantic features to produce coarse yet semantically robust CAMs, capturing salient response regions of land-cover objects in remote sensing images. FQA leverages high-resolution visual features together with class-level textual query information to enhance the activation of small-scale and fine-grained target regions.

in remote sensing images. Unlike traditional Grad-CAM methods, CSA introduces cross-modal semantic alignment to enhance category relevance by incorporating image-level textual labels, thereby improving both localization accuracy and class consistency. Specifically, let the final Transformer block's LayerNorm layer in the CLIP image encoder serve as the target layer. The last-layer feature map from the image encoder $f_L^I \in \mathbb{R}^{(h\times w)\times d_l}$ is feed into the target normalization layer yields the target feature representation $\mathcal{E}^I_{\text{target}} \in \mathbb{R}^{(h\times w)\times d}$. Then, we perform global average pooling on $\mathcal{E}^I_{\text{target}}$ to obtain the global semantic vector $\bar{\mathcal{E}}^I_{\text{target}}$, and compute a cost volume between it and the corresponding textual features $\mathcal{E}_I^{\text{T}} \in \mathbb{R}^{C_I\times d}$ for each of the C_I image-level label. Here, the classification prediction scores for the image–text pairs obtained after softmax are further supervised using the cross-entropy loss function

$$\mathcal{L}_{\text{cam}} = -\sum_{c\in C_I} y_c \cdot \log\left(\text{soft}\max\left(\mathcal{M}\left(\bar{\mathcal{E}}^I_{\text{target}}, \mathcal{E}_I^{\text{T}}\right)\right)\right) \tag{7}$$

where y_c is the one-hot encoded label vector. To generate class-specific activation maps, we backpropagate the classification loss to the target feature layer to obtain channelwise importance weights ω_c^k. The CAM for class $c \in C_I$ is then computed as

$$\mathcal{A}_c^{\text{CSA}} = \text{RELU}\left(\sum_{k=1}^{d} \omega_c^k \cdot \mathcal{E}^I_{\text{target}}(:,k)\right). \tag{8}$$

Notably, due to complex backgrounds and uneven object responses in remote sensing images, we explicitly construct a background activation map without requiring background labels. This is achieved by identifying low-activation regions across all foreground CAMs and assigning them as potential high-confidence background areas. This strategy enhances foreground–background separability and fills semantic gaps in pseudo labels, improving boundary clarity and segmentation robustness. Finally, all activation maps are passed through a contextual feature refinement (CFR) module for spatial smoothing and structural enhancement. They are then upsampled to match the original image resolution $H \times W$, producing the final coarse-grained activation tensor $\mathcal{A}^{\text{CSA}} \in \mathbb{R}^{C_I\times H\times W}$.

2) Fine-Grained Query Activation: While CSA can reliably capture salient regions of remote sensing targets, its reliance on low-resolution, deep semantic features makes it prone to missing small-scale objects and fine structural details. To complement CSA, we design the FQA module, which leverages high-resolution visual features and category-level textual guidance to enhance the activation of small-scale and fine-grained object regions, enabling more accurate and semantically aligned localization. Specifically, FQA begins by performing spatial upsampling on the main image encoding features $\mathcal{E}^I \in \mathbb{R}^{(h\times w)\times d}$ to restore detailed spatial structure. A pretrained upsampling model (SimFeatUp [9]) $\Lambda^{\mathcal{S}}$ is applied to recover the visual feature at the original image resolution

$$\mathcal{E}^I_{\text{up}} = \Lambda^{\mathcal{S}}\left(\mathcal{E}^I\right) \in \mathbb{R}^{(H\times W)\times d}. \tag{9}$$

These high-resolution visual features retain semantic context while preserving fine spatial details, thus laying a strong foundation for precise target localization. FQA then introduces the textual features $\mathcal{E}_I^{\text{T}} \in \mathbb{R}^{C_I\times d}$ corresponding to the image-level category labels as a query set. Through a cross-modal semantic interaction mechanism, it performs query-to-pixel matching, guiding the visual features to focus on regions that are semantically aligned with each category. The activation is computed as

$$\mathcal{A}^{\text{FQA}} = \text{soft}\max\left(\mathcal{M}\left(\mathcal{E}^I_{\text{up}}, \mathcal{E}_I^{\text{T}}\right)\right). \tag{10}$$

$\mathcal{A}^{\text{FQA}} \in \mathbb{R}^{C_I\times H\times W}$. Spatial locations with weak activation across all foreground categories are explicitly considered as background, similar to CSA. This semantic distribution map effectively localizes category-relevant regions across the image space, particularly benefiting the detection of small objects and object boundaries. By filling in the missing or incomplete regions left by CSA, FQA significantly improves the completeness and semantic consistency of the generated pseudo-activation maps.

3) Confidence-Aware Attention Refinement: To better adapt to the varying response characteristics of objects at different scales in remote sensing images, we adopt a confidence-adaptive fusion strategy to integrate the activation maps from the aforementioned CSA and FQA modules. This fusion aims to further improve the accuracy and structural completeness of the CAMs. Specifically, considering that the response reliability varies spatially across different categories, we perform a pixel-wise adaptive weighted fusion based on the response strength from both sources

$$\mathcal{A}_f = \frac{\gamma_{\text{CSA}}\cdot\mathcal{A}^{\text{CSA}} + \gamma_{\text{FQA}}\cdot\mathcal{A}^{\text{FQA}}}{\gamma_{\text{CSA}} + \gamma_{\text{FQA}}} \tag{11}$$

where γ_{CSA} and γ_{FQA} are confidence weights for the CSA and FQA branches, respectively, determined adaptively based on the maximum response values per category at each spatial position. This fusion mechanism allows the model to dynamically adjust the information source based on category and location, resulting in a final fused activation map $\mathcal{A}_f \in \mathbb{R}^{C_I\times H\times W}$ with enhanced multiscale perception and semantic focus.

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Despite the capability of CSA and FQA to provide class-guided pseudo-supervision, both are essentially nonlearnable and rely solely on image-level labels, limiting their ability to capture semantic consistency across target categories, especially under complex remote sensing backgrounds. This issue is further exacerbated in open-vocabulary settings, where pretrained vision-language models typically align semantics only at the image level and lack fine-grained, pixel-level modeling capability. To address these limitations and further improve the structural coherence and boundary clarity of the activation maps, we introduce a self-attention refinement mechanism based on the reshaped category-level semantic features $\tilde{\mathcal{Z}}$ during the later training stages. Specifically, we compute a spatial semantic attention map as

$$\mathcal{R}_f = \text{soft max}\left(\tilde{\mathcal{Z}} \cdot \tilde{\mathcal{Z}}^{\mathrm{T}}\right) \in \mathbb{R}^{(hw)\times(hw)}. \tag{12}$$

This attention matrix explicitly models the semantic similarity across different spatial positions, reflecting both intraobject consistency and intercategory relations. Subsequently, we apply a sparse attention propagation over the fused activation map, guided by the semantic attention map $\mathcal{R}_f$, to refine the distribution of activation responses

$$\tilde{\mathcal{A}}_f(c,:) = \mathcal{R}_f \cdot \mathcal{A}_f(c,:)^{\mathrm{T}}. \tag{13}$$

Here, each category $c \in C_I$ undergoes intrastructure attention refinement to yield the final activation map $\tilde{\mathcal{A}}_f \in \mathbb{R}^{C_I\times H\times W}$. To avoid background noise propagation, this refinement is restricted to high-confidence foreground regions, allowing attention to propagate only within reliable activation areas. Through this mechanism, the model achieves stronger intra-class response consistency, effectively mitigating spatial misalignment caused by weak vision-language coupling, and ultimately enhancing the structural integrity and class discrimination of the generated pseudo-labels.

D. Training and Inference

1) Training With Pseudo Labels: During training, the CAMs produced by the proposed TAAM are converted into pseudo labels by assigning each spatial position to the category with the maximum response, subject to a confidence threshold. Formally, the pseudo label map is defined as

$$\mathcal{Y}_{\text{pseudo}} = \begin{cases} \arg\max\limits_{c\in C_I} \tilde{\mathcal{A}}_f(c,:), & \text{if } \max_c \tilde{\mathcal{A}}_f(c,:) \geq \tau \\ \text{ignore}, & \text{otherwise} \end{cases} \tag{14}$$

where τ denotes the confidence threshold, and C_I is the set of image-level labels associated with the current image. The resulting pseudo label map $\mathcal{Y}_{\text{pseudo}} \in \mathbb{R}^{H\times W}$ is then transformed into one-hot encoding, aligned with the decoder output prediction $\hat{\mathcal{Y}}$, and used to compute the pixel-wise binary cross-entropy loss

$$\begin{aligned}\mathcal{L}_{\text{seg}} = &-\frac{1}{|\Omega|}\sum_{(i,j)\in\Omega}\sum_{c\in C}\Big[\mathcal{Y}_{\text{pseudo}}^{(i,j,c)} \cdot \log\sigma\left(\hat{\mathcal{Y}}^{(i,j,c)}\right) \\ &+ \left(1-\mathcal{Y}_{\text{pseudo}}^{(i,j,c)}\right)\cdot\log\left(1-\sigma\left(\hat{\mathcal{Y}}^{(i,j,c)}\right)\right)\Big].\end{aligned} \tag{15}$$

Here, $\hat{\mathcal{Y}}^{(i,j,c)}$ denotes the predicted logit for class $c \in C$ at pixel location (i, j), $\mathcal{Y}_{\text{pseudo}}^{(i,j,c)}$ is the one-hot encoded pseudo label, σ is the Sigmoid activation, and Ω represents the set of valid (nonignore) pixel positions.

2) Open-Vocabulary Inference: The proposed model exhibits strong open-vocabulary generalization during inference. Thanks to the text-guided semantic modeling design, the model can flexibly incorporate arbitrary category text labels without requiring additional retraining. Specifically, during inference, the test image is paired with a set of desired category-level textual descriptions and fed into the model. Through joint feature encoding and semantic decoding, the model directly outputs the pixel-level semantic segmentation results for the specified target categories. This design allows the model to generalize to novel or unseen classes, making it particularly suited for remote sensing scenarios where class vocabularies may evolve over time or vary across tasks.

IV. Experiments

A. Experimental Datasets

We conduct experiments using the LandDiscover50K dataset [36], which is specifically constructed for OVSS in remote sensing. To comprehensively evaluate the generalization capability of the proposed method across various scenarios and segmentation tasks, we further benchmark on four representative remote sensing datasets curated alongside LandDiscover50K as follows.

1) *FLAIR [37]:* Covers a wide range of land cover types, suitable for assessing model performance on large-scale semantic segmentation of diverse geographic objects.
2) *FAST [38]:* Focuses on fine-grained object segmentation, providing a challenging benchmark for evaluating the model's ability to perceive small-scale targets and fine class boundaries.
3) *ISPRS Potsdam [39]:* Shares a similar category set with LandDiscover50K, enabling the evaluation of model performance on in-vocabulary categories.
4) *FloodNet [40]:* Specializes in post-disaster flood scene understanding, making it a valuable dataset for assessing the model's transferability and robustness in extreme conditions.

Detailed statistics and category coverage for each dataset are summarized in Table I, where synonyms within a class are denoted by curly brackets {} to reflect their semantic equivalence in textual supervision.

B. Experimental Setting

All experiments are conducted on the Detectron2[1] framework for training and evaluation. We adopt the CLIP model as the backbone for vision-language feature extraction, leveraging its cross-modal alignment capabilities learned from large-scale pretraining on image–text pairs. On the text side, class names are converted into prompts using the OpenAI ImageNet-style template ("A photograph of a [CLASS]") to ensure compatibility with CLIP's training format and semantic structure. For

[1] https://github.com/facebookresearch/detectron2

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TABLE I
SUPERVISED CATEGORY VOCABULARIES IN DIFFERENT DATASETS

	Dataset	Data Volume	Supervised Category Vocabularies
Training	LandDiscover50K	51,846	'background', 'bare land', 'grass', 'pavement', 'road', 'tree', 'water', 'agriculture land', 'buildings', 'forest land', 'barren land', 'urban land', 'large-vehicle', 'swimming-pool', 'helicopter', 'bridge', 'plane', 'ship', 'soccer-ball-field', 'basketball-court', 'ground-track-field', 'small-vehicle', 'baseball-diamond', 'tennis-court', 'roundabout', 'storage-tank', 'harbor', 'container-crane', 'airport', 'helipad', 'chimney', 'expressway service area', 'expresswalltoll station', 'dam', 'golf field', 'overpass', 'stadium', 'train station', 'vehicle', 'windmill'
Testing	FLAIR	15,700	'building', 'pervious surface', 'impervious surface', 'bare soil', 'water', 'coniferous', 'deciduous', 'brushwood', 'vineyard', 'herbaceous vegetation', 'agricultural land', 'plowed land'
	FAST	3,207	'a220', 'a321', 'a330', 'a350', 'arj21', 'baseball-field', 'basketball-court', 'boeing737', 'boeing747', 'boeing777', 'boeing787', 'bridge', 'bus', 'c919', 'cargo-truck', 'dry-cargo-ship', 'dump-truck', 'engineering-ship', 'excavator', 'fishing-boat', 'football-field', 'intersection', 'liquid-cargo-ship', 'motorboat', 'other-airplane', 'other-ship', 'other-vehicle', 'passenger-ship', 'roundabout', 'small-car', 'tennis-court', 'tractor', 'trailer', 'truck-tractor', 'tugboat', 'van', 'warship'
	ISPRS Potsdam	5,328	{'road', 'parking lot'}, 'building', 'low vegetation', 'tree', 'car', {'clutter', 'background'}
	FloodNet	898	'background', 'building-flooded', 'building-non-flooded', 'road-flooded', 'road-non-flooded', 'water', 'tree', 'vehicle', 'pool', 'grass'

the visual input, remote sensing images are randomly cropped and resized to a fixed resolution of 384 × 384 to standardize training. The model is optimized using AdamW with an initial learning rate of 0.0001.

To comprehensively evaluate the model's performance in the OVSS tasks, we adopt mean intersection over union (mIoU) as the primary metric for class-level segmentation performance, and overall accuracy (OA) as a complementary metric to assess the model's ability to preserve semantic consistency and distinguish categories at the image level. This dual-metric evaluation provides a holistic understanding of the model's performance in open-world recognition and segmentation tasks.

C. Comparative Experiments and Results Analysis

The proposed method focuses on training an OVSS model tailored for remote sensing images under text-weak supervision. However, open-vocabulary methods in the remote sensing domain are still in their infancy. Relevant research remains scarce, and there is a lack of methods that are both designed for remote sensing scenarios and trained with text-weak supervision. Therefore, to comprehensively evaluate the performance of the proposed model, we compare it against several representative OVSS methods. These include the general vision-language model CLIP [7], as well as a series of recent text weakly supervised open-vocabulary segmentation methods, such as GroupViT[2] [13], MaskCLIP[3] [18], SegCLIP[4] [17], and TCL[5] [8]. In addition, we include SegEarth-OV[6] [9] as a baseline, which is one of the few existing methods specifically designed for remote sensing OVSS tasks. Notably, SegEarth-OV leverages a self-supervised pretraining strategy and does not rely on textual supervision signals. Regarding the vision-language encoder, all methods except GroupViT adopt the pretrained CLIP model as the feature extractor to ensure consistency in the visual-semantic embedding space. Among these methods, MaskCLIP is the only one that performs open-vocabulary recognition purely based on knowledge acquired during pretraining, without any fine-tuning on downstream datasets. In contrast, the other approaches perform further training on task-specific datasets to improve adaptability and segmentation accuracy under open-vocabulary settings. All comparison methods are evaluated using publicly available pretrained weights to ensure fairness and reproducibility.

1) Quantitative Evaluation: Tables II and III present the open-vocabulary recognition performance on four validation datasets. As shown, the proposed T2ASeg consistently achieves the highest mIoU across all four datasets and also obtains the best OA on three of them, demonstrating superior and comprehensive recognition performance.

Compared with existing methods, the core advantage of T2ASeg lies in its use of a remote-sensing-specific training set, LandDiscover50K, along with a carefully designed decoding enhancement mechanism and pseudo-label generation strategy. These components jointly enhance the model's generalization capability and spatial understanding in open-vocabulary recognition scenarios. In contrast, methods such as CLIP, GroupViT, MaskCLIP, SegCLIP, and TCL are all trained on natural image datasets without considering the semantic distribution and scale characteristics specific to remote sensing. Consequently, they exhibit poor performance across all datasets, with average mIoU generally below 26%, and particularly low scores for CLIP and GroupViT (below 12%). This highlights the significant domain gap when directly transferring vision-language models trained on natural images to remote sensing tasks. SegEarth-OV, one of the few open-vocabulary methods trained on remote sensing data, shows a notable performance improvement over natural-image-based methods. It even slightly outperforms T2ASeg in OA on the FloodNet dataset, indicating that its self-supervised pretraining strategy provides advantages in disaster-related scenarios. However, in terms of mIoU, T2ASeg still maintains the highest score on this dataset, reflecting stronger structural parsing capability for open-vocabulary regions.

[2]https://github.com/NVlabs/GroupViT
[3]https://github.com/chongzhou96/MaskCLIP
[4]https://github.com/ArrowLuo/SegCLIP
[5]https://github.com/khanrc/tcl
[6]https://github.com/likyoo/SegEarth-OV

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TABLE II
OPEN-VOCABULARY RECOGNITION PERFORMANCE (MIOU %) ON FOUR BENCHMARK VALIDATION DATASETS

Methods	Pre-trained Model	Training Data (Data Volume)	Downstream Validation Datasets				Average
			FLAIR	FAST	ISPRS Potsdam	FloodNet	
CLIP [7]	✗	WIT(400M)	11.11	2.82	16.55	10.63	10.28
GroupViT [13]	✗	CC(12M)+YFCC(14M)	9.31	5.76	22.26	10.06	11.85
MaskCLIP [18]	CLIP-ViT-B/16	✗	9.33	7.56	29.01	21.52	16.86
SegCLIP [17]	CLIP-ViT-B/16	CC(403M)+COCO(400K)	9.34	9.66	30.87	22.75	18.16
TCL [8]	CLIP-ViT-B/16	CC(3M)+CC(12M)	16.74	12.59	41.93	29.23	25.12
SegEarth-OV [9]	CLIP-ViT-B/16	Million-AID(16K)	16.74	13.07	47.19	33.40	27.60
T2ASeg	CLIP-ViT-B/16	LandDiscover50K(50K)	**19.62**	**15.04**	**48.28**	**35.63**	**29.64**

TABLE III
OPEN-VOCABULARY RECOGNITION PERFORMANCE (OA %) ON FOUR BENCHMARK VALIDATION DATASETS

Methods	Pre-trained Model	Training Data (Data Volume)	Downstream Validation Datasets				Average
			FLAIR	FAST	ISPRS Potsdam	FloodNet	
CLIP [7]	✗	WIT(400M)	24.88	6.68	26.64	16.24	18.61
GroupViT [13]	✗	CC(12M)+YFCC(14M)	20.79	12.75	35.29	27.63	24.12
MaskCLIP [18]	CLIP-ViT-B/16	✗	16.25	14.62	49.45	40.66	30.25
SegCLIP [17]	CLIP-ViT-B/16	CC(403M)+COCO(400K)	17.34	19.58	50.39	42.95	32.57
TCL [8]	CLIP-ViT-B/16	CC(3M)+CC(12M)	32.17	22.62	56.60	54.45	41.46
SegEarth-OV [9]	CLIP-ViT-B/16	Million-AID(16K)	37.46	22.99	66.55	**59.86**	46.72
T2ASeg	CLIP-ViT-B/16	LandDiscover50K(50K)	**41.99**	**25.13**	**70.58**	54.54	**48.06**

From a dataset-specific perspective, T2ASeg demonstrates excellent adaptability across all four scenes. On FLAIR, which contains a wide range of land cover types, T2ASeg achieves the highest accuracy thanks to its robust open-vocabulary semantic modeling. On FAST, which emphasizes small objects and fine-grained categories, the structure-aware design of T2ASeg delivers a clear advantage. On ISPRS Potsdam, where categories align with those in the training set, T2ASeg achieves record-high accuracy for seen classes. Even in disaster scenarios such as FloodNet, it maintains the strongest mIoU, reflecting its ability to parse complex semantic structures. In summary, T2ASeg not only outperforms existing methods in overall metrics, but more importantly, its training strategy and architectural design are well-aligned with the semantic complexity and openness inherent to remote sensing tasks. It consistently delivers robust performance across datasets of different scenes, granularities, and label distributions. These results strongly validate the effectiveness and generalizability of the proposed text-weakly supervised training mechanism and structure-semantic enhancement framework for OVSS in remote sensing.

2) Qualitative Evaluation: To further validate the open-vocabulary recognition capability of the proposed model in real-world remote sensing imagery, we conducted qualitative comparisons on representative sample images from the four validation datasets. The compared methods include TCL (a strong performer among natural image-trained models), SegEarth-OV (a remote sensing-trained method), and our proposed T2ASeg. The results are illustrated in Fig. 4.

Experimental observations show that TCL generally performs the worst across all scenes. For instance, in the FloodNet disaster scenario, TCL frequently misclassifies water-logged areas as grass, and its predictions between "water" and "grass" are unstable, indicating a poor adaptation to the semantic structures of remote sensing scenes.

In scenes involving large objects or ambiguous boundaries (e.g., FLAIR and FloodNet), SegEarth-OV benefits from remote sensing-specific training data and improves recognition accuracy overall. However, it still struggles with modeling complex category boundaries. For example, in FLAIR, it often confuses "coniferous" and "bare soil" due to blurred segmentation borders. In contrast, T2ASeg demonstrates superior boundary awareness and semantic consistency, clearly distinguishing between semantically similar land cover types and showing strong structural parsing ability and classification stability.

In scenes with small objects and fine-grained categories, such as ISPRS Potsdam and FAST, both SegEarth-OV and T2ASeg perform relatively well, yet T2ASeg maintains a clear edge. For example, in ISPRS Potsdam, while SegEarth-OV can generally recognize "low vegetation," its handling of boundaries with small categories like "car" remains coarse. In FAST, SegEarth-OV achieves stable segmentation for vehicle classes, but T2ASeg exhibits significantly better performance in fine-grained ship categories (e.g., "fishing-boat," "dry-cargo-ship"), indicating a stronger ability to model fine-grained category distinctions.

Overall, T2ASeg consistently demonstrates superior structural reconstruction and semantic discrimination across diverse scenarios, particularly in large-object boundary delineation, semantically similar class separation, and small-object fine-grained modeling. These advantages stem from its structure-semantic joint enhancement mechanism, which improves spatial perception, and its semantic alignment training strategy, which is tailored to remote sensing category characteristics. The qualitative results further

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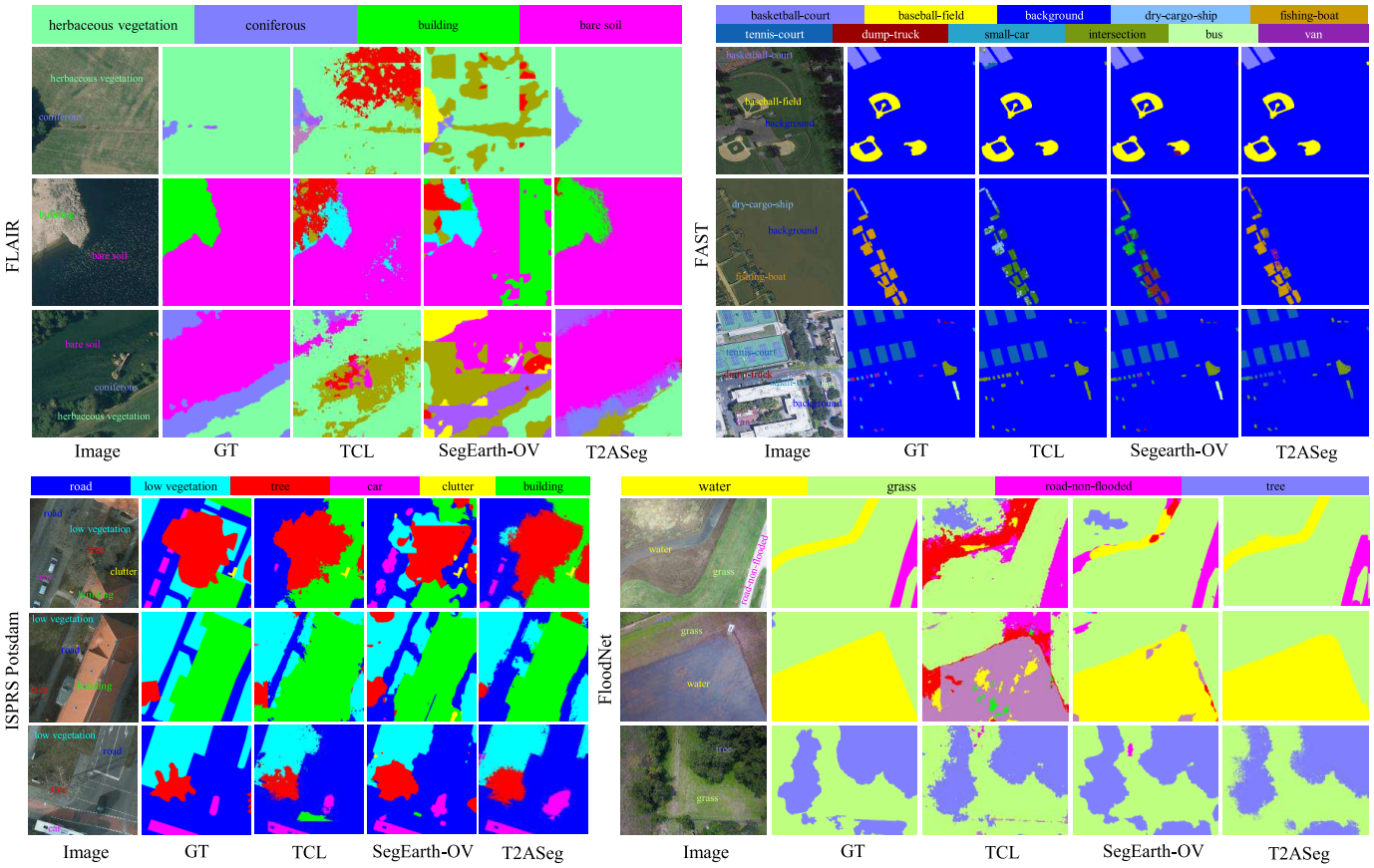


Fig. 4. Qualitative comparison between different OVSS methods on the validation datasets. The compared methods include TCL, representing a strong performer among natural image-trained models; SegEarth-OV, a representative remote sensing-trained method; and the proposed T2ASeg.

TABLE IV
ABLATION RESULTS (MIOU %) ON MSSEM

Methods	FLAIR	FAST	ISPRS Potsdam	FloodNet	Average
w/o SE	19.16	14.41	46.12	35.48	28.79
w/o CE	18.85	14.73	47.10	34.84	28.88
w/o MSSEM	18.25	13.93	45.61	34.46	28.06
T2ASeg	**19.62**	**15.04**	**48.28**	**35.63**	**29.64**

confirm the effectiveness and generalizability of T2ASeg in open-vocabulary remote sensing semantic segmentation tasks.

V. DISCUSSION

A. Ablation Study

The proposed T2ASeg framework comprises two key modules: the MSSEM in the semantic decoding and the TAAM in the pseudo-label generating. To validate the effectiveness and contributions of each module within the overall framework, we conduct a series of ablation experiments from two perspectives.

1) Ablation Study On MSSEM: To further evaluate the effectiveness of the MSSEM module in the semantic decoding stage, Table IV reports the results under different ablation settings, including removing the structural enhancement sub-module ("w/o SE"), removing the class-wise enhancement sub-module ("w/o CE"), completely removing the entire MSSEM module ("w/o MSSEM"), as well as the full T2ASeg with MSSEM. The results show that when the MSSEM module is entirely removed, the model performance drops significantly across all four datasets, yielding the lowest OA, which verifies the critical role of MSSEM in semantic decoding. A closer analysis reveals that removing the structural enhancement leads to a notable decline in datasets with clear object boundaries, such as ISPRS Potsdam, indicating that structural enhancement is essential for capturing spatial details and boundary delineation. In contrast, removing the class-wise enhancement causes greater degradation on the FloodNet dataset, highlighting its importance in improving intra-class consistency and reducing interclass confusion under complex post-flood scenes. Ultimately, the synergy of structural and class-wise enhancements enables the complete T2ASeg to achieve the best average performance across all datasets.

2) Ablation Study On TAAM: TAAM mainly consists of two class activation submodules and one confidence-aware refinement submodule, which are responsible for generating response regions and refining confidence in the pseudo-labels. To analyze the individual contributions of the TAAM submodules in detail, we design ablation experiments with different submodule configurations. Specifically, in Table V, "w/ CSA" denotes using only the CSA submodule to generate CAMs for pseudo-labeling; "w/ FQA" uses only the FQA submodule; and "w/o CAR" uses both CSA and FQA for activation fusion but excludes the CAR self-supervised refinement mechanism.

From the results in Table V, it is evident that the different submodules contribute distinctively to pseudo-label generation. The "w/ CSA" setting achieves the lowest average mIoU,

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TABLE V
ABLATION RESULTS (MIOU %) ON TAAM

Methods	FLAIR	FAST	ISPRS Potsdam	FloodNet	Average
w/ CSA	15.42	12.68	38.83	32.68	24.90
w/ FQA	18.42	14.20	41.66	32.58	26.72
w/o CAR	19.31	14.81	44.75	34.89	28.44
T2ASeg	**19.62**	**15.04**	**48.28**	**35.63**	**29.64**

as the CSA-based CAMs tend to focus on highly salient regions with high localization accuracy but often fail to capture the complete structure of the target, leading to incomplete pseudo-labels. In contrast, the "w/ FQA" setting improves the average mIoU to 26.72%. By leveraging a semantic similarity-guided query mechanism, FQA generates CAMs that better cover the full extent of target regions. However, it is also prone to introducing noise and false activations, which may degrade pseudo-label quality. The "w/o CAR" setting, which fuses both CSA and FQA outputs, achieves a higher average mIoU, demonstrating the complementary nature of the two activation strategies in terms of spatial scope and semantic relevance. The fused CAMs strike a better balance between accuracy and coverage. Finally, the proposed T2ASeg model, which incorporates the complete CAR module, achieves the highest performance with an average mIoU of 29.64%, validating the effectiveness of CAR in refining the fused activation maps. By introducing a self-attention mechanism over visual features, CAR adaptively optimizes the initially fused response regions, suppresses noise, and enhances structural coherence and boundary quality of the pseudo-labels, thus playing a key role in boosting overall segmentation performance.

B. Visualization Analysis

1) Visualization of Attention Enhancement via MSSEM: To further validate the effectiveness of the proposed MSSEM module in enhancing model representation under text-based weak supervision, we present a visual comparison of attention weight distributions over target land-cover objects in remote sensing images, before and after incorporating the MSSEM module. Representative examples are shown in Fig. 5.

As observed, without the MSSEM module ("w/o MSSEM"), the model exhibits dispersed attention distributions over land-cover categories, often resulting in missed detections or false activations. Moreover, it is highly sensitive to scene variations; for instance, the model tends to lose target information when detecting "plane" objects under strong illumination or "baseball-court" objects in low-contrast background conditions. In contrast, after introducing the MSSEM module ("w/ MSSEM"), the model benefits from enhanced spatial structure modeling and category-level semantic modeling. This leads to more concentrated attention on target regions and sharper boundary delineation, significantly improving both spatial localization and category discrimination. In addition, the model demonstrates greater robustness to scene variations, maintaining consistent performance across diverse imaging conditions. Overall, the MSSEM module substantially boosts the model's robustness and generalization capability in open-vocabulary remote sensing semantic segmentation tasks.

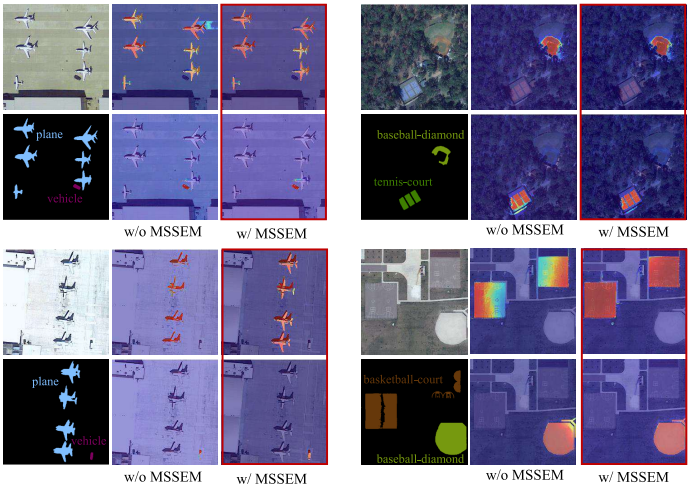


Fig. 5. Visual comparison of attention distribution before and after applying the MSSEM module. The examples include various land-cover categories (e.g., "plane," "vehicle," "baseball-court") under diverse scene conditions.

2) Visualization of Class Activation and Pseudo-Label Results: Fig. 6 presents the CAMs and corresponding pseudo-labels generated by the CSA method, FQA method, and the proposed TAAM method. A comparative analysis highlights differences in activation scope and accuracy across methods. The CSA method mainly relies on salient activation mechanisms to generate response maps, which are capable of accurately locating the core areas of targets. High-confidence responses (red regions) often focus on the central or most prominent parts of the object, resulting in relatively accurate class judgments. However, the activation maps generated by CSA tend to have limited coverage, leading to pseudo-labels with fragmented regions and incomplete object structures. The FQA method, on the other hand, models semantic associations between features, enhancing the overall responsiveness of object regions. Its activation maps tend to cover larger portions of the target, offering improved structural completeness over CSA. However, lacking saliency constraints during semantic feature aggregation, FQA can mistakenly activate background regions that are semantically similar to the target. This introduces semantic confusion between object and nonobject classes, thereby compromising pseudo-label accuracy.

In contrast, the TAAM method combines the salient localization capabilities of CSA with the semantic structural modeling strength of FQA, and further incorporates a confidence-aware self-attention refinement mechanism based on visual features. This approach strengthens the intra-class response consistency and balances activation across the entire object area. Specifically, TAAM achieves more comprehensive coverage of target regions while effectively suppressing background interference through fine-grained constraints. As a result, the activation maps demonstrate superior performance in terms of semantic coverage, boundary continuity, and response uniformity. The final pseudo-labels generated by TAAM strike a well-rounded balance between accuracy and completeness, closely aligning with the structure and semantic distribution of ground-truth labels and significantly enhancing supervision quality for downstream segmentation.

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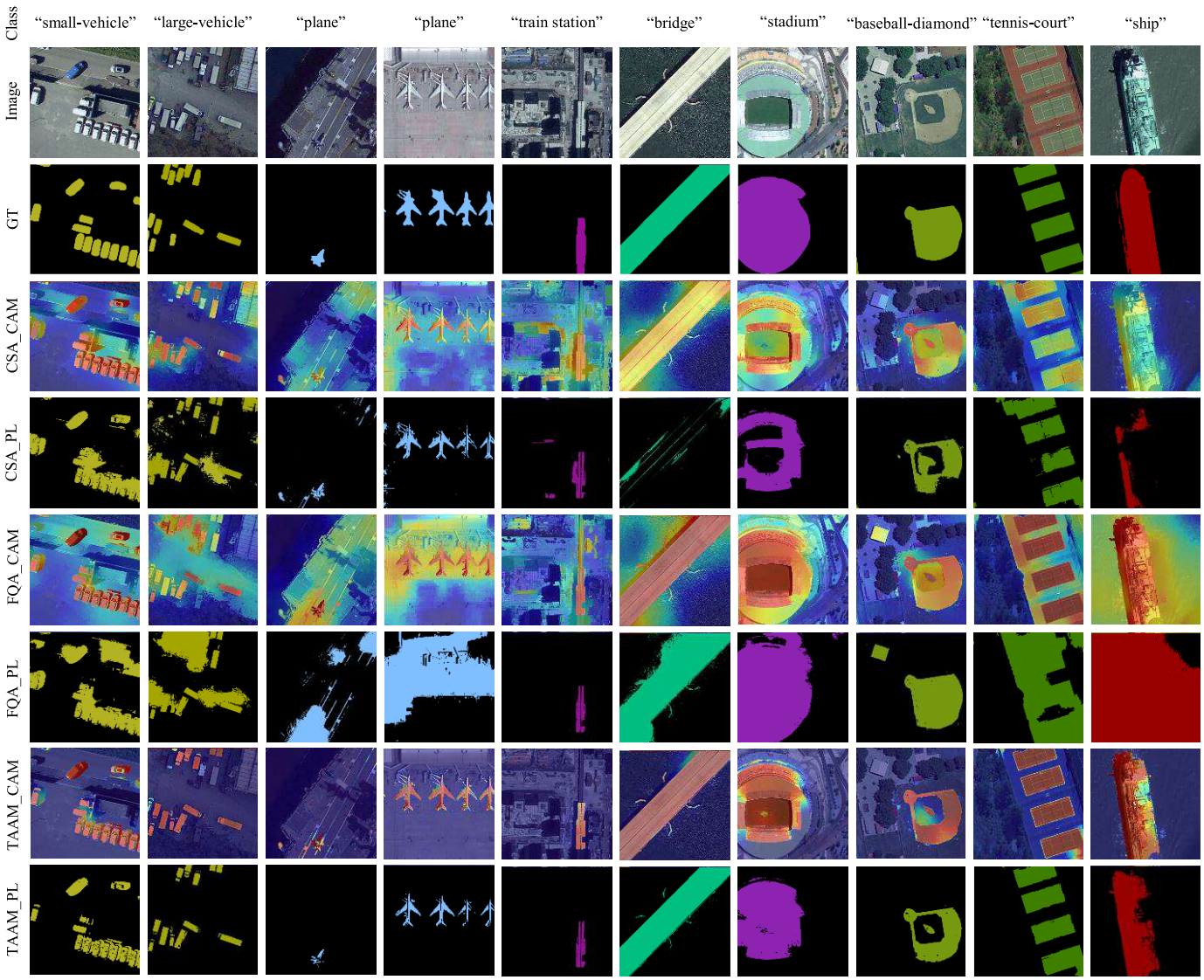


Fig. 6. Visualization of CAMs and pseudo-labels generated by CSA, FQA, and TAAM. CSA focuses on salient key regions but lacks structural integrity; FQA enhances overall coverage yet introduces background confusion; TAAM achieves a better balance by integrating saliency localization and semantic structure modeling.

C. Hyperparameter Analysis

1) Multiscale Window Selection in MSSEM: In the MSSEM, we introduce multiscale windows to enhance the modeling of local spatial structures, thereby improving the model's ability to perceive multiscale land-cover patterns in remote sensing images. In this experiment, the commonly used window size $(h/2, w/2)$ in the Swin Transformer Block is adopted as the baseline, and three auxiliary windows of different scales are incorporated in combination to investigate their impact on semantic segmentation performance.

Fig. 7 presents the results of different window configurations on four validation datasets. It can be observed that, compared with the single baseline window, introducing auxiliary windows consistently improves the open-vocabulary segmentation performance across all datasets. Among them, the dual-scale combination "$(h/2, w/2) + (h/6, w/6)$" achieves the most significant improvement, indicating that smaller scale windows are more effective in adapting to complex and diverse remote sensing scenes, thereby enhancing the model's generalization capability.

2) Confidence Threshold Selection in CAR: To generate high-quality pseudo-labels under text-weak supervision, we construct supervision signals based on CAMs to guide the decoder in learning semantic segmentation. However, due to the inherent noise and uncertainty in weakly supervised activation maps, directly using raw activation outputs may propagate errors in pseudo-labels and degrade model performance. To address this, we introduce a confidence threshold mechanism during pseudo-label generation. This filters out low-confidence regions, aiming to strike a balance between label completeness and label accuracy. To explore the influence of the threshold on model performance, we conduct an additional hyperparameter experiment by testing different threshold values.

Fig. 8 shows the mIoU results of model performance under different confidence threshold settings. From the exper-

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TABLE VI

FINE-GRAINED SEGMENTATION PERFORMANCE COMPARISON ON SEMANTICALLY CONFUSING CATEGORIES (MIOU %)

Methods		CLIP [7]	GroupViT [13]	MaskCLIP [18]	SegCLIP [17]	TCL [8]	SegEarth-OV [9]	T2ASeg
surface	pervious surface	4.07	1.01	5.88	6.38	4.21	4.65	**7.32**
	impervious surface	**5.48**	0.11	4.06	3.42	3.66	0.04	1.31
	mIoU	*4.78*	*0.56*	***4.97***	*4.90*	*3.94*	*2.35*	*4.32*
land	agricultural land	11.09	9.94	10.12	13.97	15.40	19.52	**33.50**
	plowed land	4.58	3.45	5.26	9.08	3.06	7.37	**10.04**
	mIoU	*7.84*	*6.70*	*7.69*	*11.53*	*9.23*	*13.45*	***21.77***
ship	dry-cargo-ship	1.84	21.34	0.50	5.79	**34.49**	0.17	8.92
	engineering-ship	1.93	2.09	0.90	4.60	**5.48**	2.69	2.17
	liquid-cargo-ship	0.51	7.82	0.40	14.40	**17.51**	0.11	3.76
	passenger-ship	0.85	0.29	0.98	**7.39**	5.72	0.35	3.13
	warship	3.04	3.80	7.88	0.00	**10.97**	10.44	9.96
	other-ship	0.85	4.94	1.09	**5.01**	0.00	1.58	0.09
	mIoU	*1.50*	*6.71*	*1.96*	*6.20*	***12.36***	*2.56*	*4.67*
building	building-flooded	4.53	**10.03**	0.14	9.56	7.32	0.01	8.04
	building-non-flooded	4.53	10.65	22.40	19.28	29.40	**39.35**	33.08
	mIoU	*4.53*	*10.34*	*11.27*	*14.42*	*18.36*	*19.68*	***20.56***
road	road-flooded	2.07	3.52	3.98	11.64	13.25	19.19	**26.59**
	road-non-flooded	4.54	5.69	29.65	18.66	32.72	**56.73**	43.35
	mIoU	*3.31*	*4.61*	*16.82*	*15.15*	*22.99*	***37.96***	*34.97*

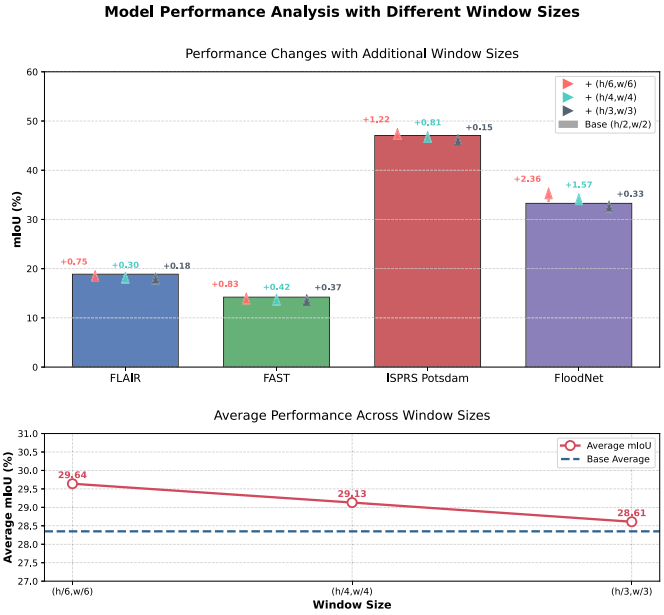


Fig. 7. Model performance analysis with different window size settings. The bar chart at the top shows the results of introducing auxiliary windows of various sizes on top of the base setting to capture the semantic information of objects at different scales. The line chart at the bottom illustrates the average performance across four test sets under different settings.

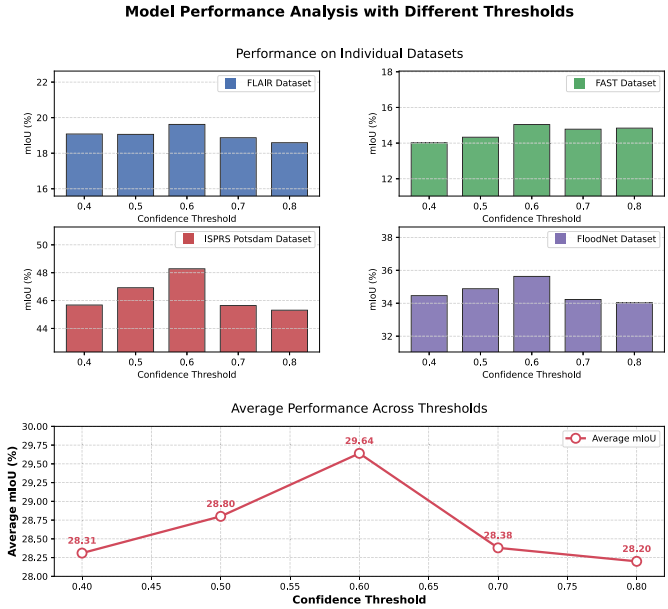


Fig. 8. Model performance analysis with different confidence threshold settings. The bar chart at the top presents the performance of four test datasets under various threshold settings, while the line chart at the bottom illustrates the average performance across the four test sets under different settings.

imental results, a low threshold introduces excessive noisy pseudo-labels, which mislead the model training and result in performance degradation. In contrast, a high threshold provides more accurate supervision but suffers from excessive omissions, leading to insufficient guidance, limited learning ability, and a tendency to overfit. Overall, the comparison indicates that when the threshold is set to 0.6, the model achieves the best balance between accuracy and completeness, yielding the optimal performance.

D. Fine-Grained Semantic Category Analysis

In open-vocabulary remote sensing semantic segmentation, pretrained models can perform pixel-level segmentation based on the category text semantics provided by arbitrary test datasets, predicting the semantic label of each pixel accordingly. However, when semantically similar or easily confusable categories exist in the dataset, the difficulty of fine-grained category discrimination increases substantially, thereby imposing higher demands on segmentation accuracy. To this end, we provide in this section a fine-grained analysis of semantically confusing categories, as shown in Table VI, and compare the strengths and limitations of different methods in handling categories with ambiguous semantic boundaries.

Table VI reports the fine-grained segmentation performance of various methods on several semantically similar categories. As observed in the surface category, all methods exhibit rela-

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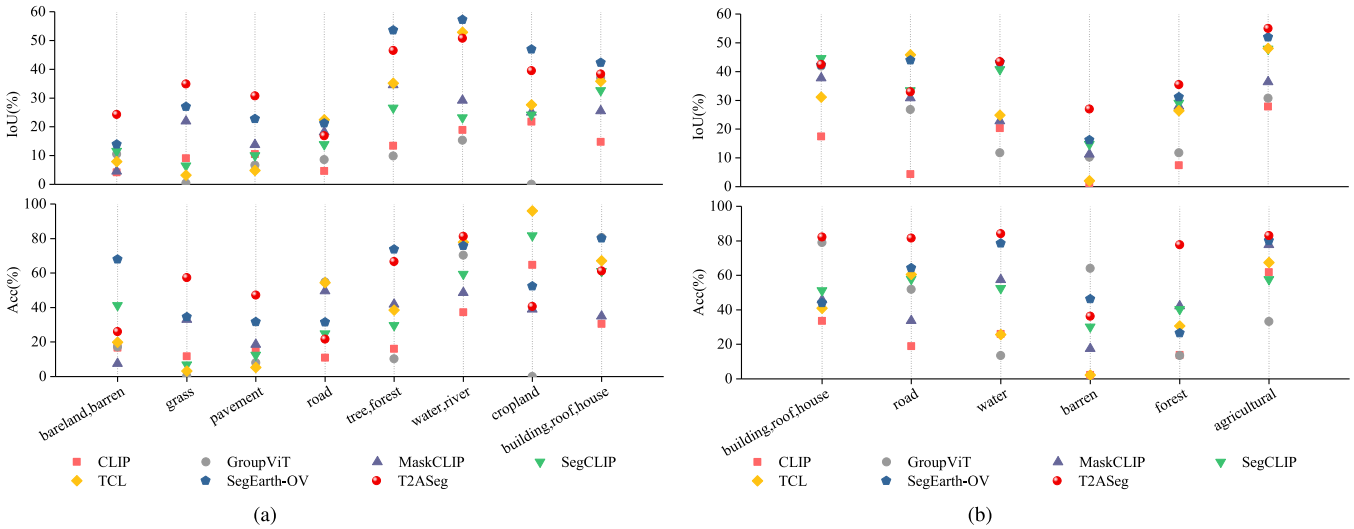


Fig. 9. Comparison of different OVSS methods on the (a) OpenEarthMap and (b) LoveDA datasets in terms of IoU and Acc. The evaluation covers diverse scenarios and categories, highlighting the applicability of each method to land-cover and urban environments.

tively weak performance in distinguishing "pervious surface" from "impervious surface," suggesting the difficulty of capturing subtle differences in surface materials. For the land category, particularly between "agricultural land" and "plowed land," the proposed T2ASeg method achieves a remarkable improvement, reaching an mIoU of 21.77%, significantly outperforming existing approaches. In the ship category, due to the high semantic and visual similarity among subclasses, most methods exhibit limited fine-grained performance with overall low mIoU; although TCL achieves better results in certain subclasses, its overall stability remains insufficient. For the building category, T2ASeg demonstrates robust performance in distinguishing "flooded" from "nonflooded" scenes, achieving an mIoU of 20.56%, comparable to SegEarth-OV. Finally, in the road category, T2ASeg shows clear advantages in both "road-flooded" and "road-nonflooded," which is significantly higher than most baselines. Overall, these results indicate that T2ASeg demonstrates stronger discriminative capability and generalization in fine-grained category distinction, effectively mitigating confusion and enhancing fine-grained segmentation performance in open-vocabulary remote sensing tasks.

It should be noted that although T2ASeg demonstrates clear advantages in several fine-grained categories, its performance is not consistently optimal across all cases. For instance, in certain ship subclasses and surface cover types, other methods, such as TCL or SegEarth-OV, still achieve competitive results on specific metrics. This indicates that T2ASeg still has room for improvement when dealing with extremely similar or boundary-ambiguous categories.

E. Applicability to Real-World Land-Cover Recognition

To further evaluate the applicability of the proposed method in real-world scenarios, we conducted additional experiments on two representative land-cover datasets: OpenEarthMap and LoveDA. The OpenEarthMap dataset is a global-scale land-cover dataset collected from multiple regions and sensors, covering a diverse range of natural and man-made classes, which provides a reliable benchmark for assessing generalization to heterogeneous environments. LoveDA dataset, on the other hand, consists of land-cover scenes from both urban and rural areas, emphasizing fine-grained recognition in complex human settlement environments.

As illustrated in Fig. 9, the overall results demonstrate that T2ASeg consistently achieves superior performance across diverse scenes and categories, with particularly strong generalization and fine-grained discriminative capability. On OpenEarthMap dataset, T2ASeg attains leading performance in both IoU and Acc across most categories, with remarkable gains in "tree/forest," "water/river," and "cropland." Notably, for semantically similar categories such as "bareland" and "cropland," existing methods often fail to yield stable predictions, whereas T2ASeg exhibits clear robustness in distinguishing such ambiguous boundaries. On LoveDA dataset, T2ASeg again shows prominent advantages in critical categories including "building/roof/house," "road," and "agricultural land." It consistently achieves the highest IoU and maintains strong accuracy, indicating its capacity to adapt to the heterogeneous land-cover patterns in complex urban–rural transitional areas. Taken together, the results across these two real-world datasets substantiate the practical applicability of T2ASeg for open-vocabulary remote sensing semantic segmentation. Beyond benchmark datasets, the method demonstrates competitive performance in realistic land-cover recognition tasks, highlighting its robustness, fine-grained discrimination, and generalization ability—thereby providing a solid foundation for deployment in practical scenarios.

Nevertheless, it is important to recognize that the performance of T2ASeg on real-world datasets also presents certain limitations. Specifically, while it generally outperforms most baselines on OpenEarthMap and LoveDA datasets, it does not consistently achieve the best results across all categories. In particular, for several highly confusing classes, the accuracy may still degrade. This suggests that the method's generalization ability under complex scenarios has room for further

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enhancement, and future work could explore the integration of multimodal priors or more refined category modeling to improve its robustness and stability.

VI. Conclusion

This article addresses two key challenges in OVSS of remote sensing imagery under text-based weak supervision: incomplete and unreliable pseudo-labels, as well as label noise that disrupts effective segmentation learning. To overcome these, we introduce T2ASeg, a unified framework that synergistically improves pseudo-label generation and semantic decoding to produce reliable labels and learn robustly from noise. Specifically, we design a text-guided class activation mechanism and an MSSEM to improve the semantic consistency and spatial integrity of pseudo labels, while also enhancing the model's generalization and discrimination capabilities in complex object structures and open-vocabulary scenarios. Extensive experiments conducted on several challenging remote sensing datasets demonstrate that T2ASeg consistently outperforms existing mainstream methods in terms of segmentation accuracy, structural completeness, and open-vocabulary recognition performance. Nevertheless, the current experiments also reveal certain limitations, such as suboptimal performance on highly confusing fine-grained categories and inconsistent generalization across complex real-world datasets. To address these challenges, future work will focus on developing more targeted and scalable OVSS approaches that leverage the unique characteristics of remote sensing imagery, thereby further enhancing robustness, fine-grained discrimination, and practical applicability.

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